Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER NETWORKS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.

(2) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) How does information get passed from one layer to the next in the OSI Reference Model ? Explain the concept of encapsulation and decapsulation. 5(CO1)

(b) Explain various Networking devices. Specify clearly the networking devices used at different layers of network model with justification. 5(CO1)

2. (a) Explain the process of connection set-up and connection tier-down in virtual circuit network. 5(CO3)

(b) List three categories of Multiple Access Protocols. Discuss any one protocol in detail. 5(CO3)

3. (a) Explain Go-Back-N ARQ protocol. Justify that Go-Back-N ARQ protocol is efficient than Stop-and-Wait ARQ protocol with reference to bandwidth utilization. 5(CO4)

(b) Discuss the relationship between STS and STM. What are the user data rates of STS-3, STS-9, and STS-12 ? Show STS-9's can be multiplexed to create an STS-36. IS there any extra overhead involved in this type of multiplexing ? 5(CO4)

4. (a) Determine the shortest path from A to E using Dijkstra Algorithm and also discuss teh scenario where Dijkstra algorithm may not work. Assume node A is a source Node.

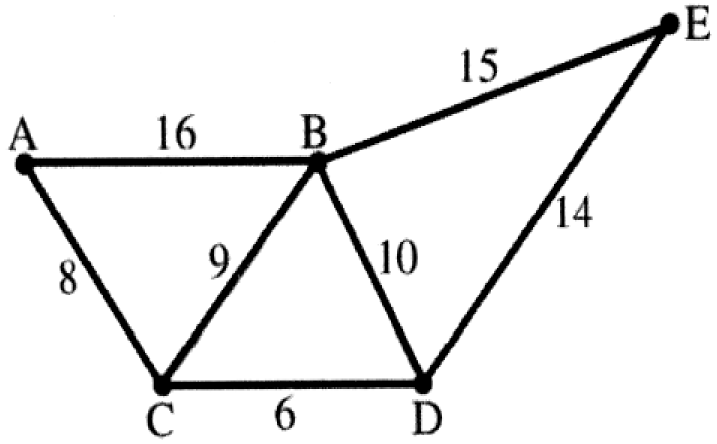


Fig. Q. 4(a)

5(CO3)

(b) An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three subnets of customers as follows :

(i) The first subnet has 64 customers; each needs 128 addresses.

(ii) The second subnet has 128 customers; each needs 128 addresses.

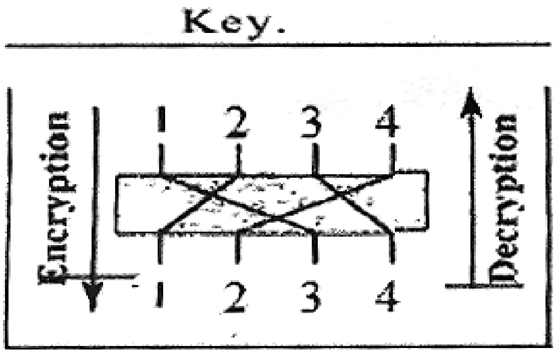
(iii) The third subnet has 128 customers; each needs 64 addresses.

Design the sub-blocks and find out how many addresses are still available after these allocations. 5(CO5)

5. (a) Differentiate between Leaky bucket and token bucket are traffic shaping techniques. 5(CO4)

(b) Digital signature provides message integrity, authentication, and non-repudiation but cannot provide confidentiality for the message. Justify. 5(CO2)

6. (a) For the key-generator shown below, Find out the key to be used to generate the cipher text.



Encrypt the message “HELLO MY DEAR,” using the above key. 5(CO2)

(b) For a RSA algorithm with p = 3 and q = 7, evaluate public key and private key. Also obtain the ciphertext for the plaintext ‘FACE’. 5(CO2)